Semantic Segmentation of Airborne LiDAR Point Clouds With Noisy Labels

Yuan Gao, Shaobo Xia, Cheng Wang, Xiaohuan Xi, Bisheng Yang, and Chou Xie

***Abstract*— High-quality point cloud annotation is labor-intensive and time-consuming, but it serves as a critical factor driving the success of LiDAR point cloud semantic segmentation. Leveraging low-quality labels in LiDAR point cloud processing is overlooked, despite the fact that noisy annotation has low labeling costs and abundant cross-modal resources (e.g., labels from images). To this end, we thoroughly investigate the performance of airborne LiDAR point cloud semantic segmentation models using noisy labels for the first time and find that it is closely related to object categories and learning stages. Then we propose a new semantic segmentation framework for LiDAR point cloud noisy learning called adaptive dynamic noise label correction (ADNLC), which consists of weak category priority, dynamic monitoring (DM), and historical choice (HC). With these methods, we can adaptively correct the noise labels of different categories according to their specific learning situations. Finally, we provide a comprehensive process for noise simulation, accuracy evaluation, and comparisons in airborne LiDAR point cloud learning from noisy labels. We conduct experiments on the ISPRS 3-D Labeling Vaihingen and Large-scale ALS data for Semantic Labeling in Dense Urban Areas (LASDU) datasets, and the results show that our ADNLC outperforms baseline methods by 30% and 16%, respectively, verifying the superiority of ADNLC and demonstrating the potential of noise labels in LiDAR data processing.**

***Index Terms*— Airborne LiDAR point cloud, noise learning, noisy labels, point cloud semantic segmentation.**

I. Introduction

LARGE-SCALE airborne LiDAR point clouds are integral to applications such as urban modeling [1], virtual reality [2], forest mapping [3], and solar energy estimation [4], among others [5]. Point cloud semantic segmentation, which assigns category labels to each individual point, serves as a fundamental process in the analysis and utilization of airborne LiDAR point clouds. Recent studies on airborne LiDAR point cloud semantic segmentation have demonstrated promising results on a number of datasets [6]. However, most methods need dense and precise point-wise annotations [2], [7], which are resource-intensive to collect, posing challenges in constructing large-scale datasets.

In recent years, there has been a growing interest in training models that solely use sparse annotations [8], [9]. For example, researchers are keen on training a semantic segmentation model using only 1% [10] or even 0.1% [11] of the labeled data. In addition, other sparse annotation strategies are also widely explored, such as subscene-level [12] and scene-level annotations [13], [14], one label per object [15], bounding box annotations [16], and brush annotations [17]. Despite these methods reducing the annotation work, the imperative for precise and perfect annotations compels professionals to utilize the software for meticulously annotating each scene or point cloud. This is not only technically demanding but also takes a lot of time and effort, resulting in a labeled dataset with a limited size.

Conversely, low-precision annotations can be obtained in various ways, such as wrong labels in manual annotations (e.g., ScanNetV2 [18]), crowd-sourced annotations [19] (e.g., OpenStreetMap [20]), and labels from pretrained models [21] (e.g., foundation models [22]). Other sources include labels transferred from imagery [23] and historical maps [24]. As a result, compromising on annotation quality can appreciably reduce costs, significantly reduce requirements for labeling techniques, and also unlock the potential of multimodal data. It is important to note that labeling errors do exist in many manually labeled datasets [18]. Nevertheless, studies on noise learning in LiDAR point clouds are very limited, and the answers to the following questions remain unclear.

1) To what extent does low-quality annotation impact airborne LiDAR point cloud semantic segmentation?
2) Is it feasible to train a comparable model to existing ones for airborne LiDAR point clouds with noisy annotations?

To this end, we conduct a comprehensive investigation into learning from noisily labeled airborne LiDAR point clouds. We evaluate the performance of point cloud learning frameworks at different noise levels. It is found that the network learns both the clean labels and the noise labels for all categories concurrently. While learning performance for noise-free labels varies across categories, an overall pattern

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emerges: the network's efficiency for learning noise-free labels diminishes over time, in contrast to an upward trend for noisy labels. Based on this, we design a robust point cloud learning framework capable of significantly mitigating the impact of noisy labels. In summary, our contributions are as follows.

1) We conduct a thorough analysis of how noisy labels affect semantic segmentation in LiDAR point clouds. We discover that the network learns both correct and incorrect labels simultaneously, but the impact of incorrect labels on the model gradually surpasses that of correct labels as the training progresses.
2) We propose a novel semantic segmentation framework for noisily labeled LiDAR point clouds. This framework employs new methods like weak category priority, dynamic monitoring (DM), and historical choice (HC), outperforming the baseline methods by a large margin.
3) As the first work on robust learning for airborne LiDAR point clouds, we establish a new benchmark, including noise simulation, method comparison, and accuracy evaluation.

The remainder of this article is organized as follows. Section II surveys prior studies in airborne LiDAR point cloud semantic segmentation, weakly supervised learning, and noise learning. Our proposed framework is described in Section III. Section IV presents comparative experiments and ablation studies. Section V concludes the work and discusses potential directions for future research.

II. Related Work

A. Semantic Segmentation in Point Clouds

1) Point Cloud Segmentation: Point cloud semantic segmentation refers to the process of partitioning the unlabeled points into subsets that share similar semantic attributes [2]. Since standard 2-D convolutional operations are designed for ordered data (e.g., 2-D images) and cannot process huge, unorganized point clouds, various networks with different structures for 3-D point clouds have emerged [25]. Among them, point-based methods have been greatly developed [18]. PointNet [26] emerges as a pioneering approach to learning features directly from point clouds using shared multilayer perceptrons and global feature aggregation via pooling functions. However, it lacks consideration for the local context. Kernel point convolution (KPConv) [27] introduces a novel convolution operator with both rigid and deformable configurations, enabling the network to adaptively learn an optimal number of kernel points, thus offering greater flexibility than traditional fixed-kernel convolutions. MinkowskiNets, as proposed by Choy et al. [28], leverage 4-D spatio-temporal convolutional neural networks with sparse tensors and generalized sparse convolution for processing high-dimensional 3-D videos. RandLA-Net [29] presents an efficient, lightweight architecture that incorporates random sampling to enhance computational and memory efficiency, along with a local feature aggregation module to retain critical feature information. The point transformer [30] efficiently processes unstructured point clouds by utilizing a local–global attention mechanism to enhance feature extraction.

2) ALS Point Cloud Segmentation: Despite recent advances in semantic segmentation of point clouds, its direct application to large-scale outdoor airborne LiDAR point clouds remains limited [31]. Huang et al. [32] developed a method for airborne LiDAR point cloud segmentation employing a hierarchical approach that begins with learning multilevel local context using PointNet++. They then apply joint manifold learning for feature embedding into a low-dimensional space, culminating in global optimization via the Markov random field algorithm. Li et al. [33] introduced a geometry-aware network that enhances segmentation outcomes through end-to-end learning, incorporating comprehensive geometric, hierarchical, and elevation information. Zhao et al. [34] utilized transfer learning to improve airborne LiDAR point cloud segmentation, achieving higher accuracy by acquiring multiscale and multiperspective feature representations. Huang et al. [35] proposed the global relation-aware attention network (GraNet), incorporating a local spatial attention convolution module (LoSDA) for local feature learning, complemented by a global relation-aware attention (GRA) module for global structural information assimilation, resulting in superior performance relative to existing state-of-the-art techniques. Li et al. [36] introduced an attentional graph gathering and merging convolution (AGGM Convolution) for the extraction of spatial geometric structural features, along with an upsampling module designed to integrate multiscale feature sets. Lin et al. [37] introduced the local and global encoder network (LGENet), which captures local geometric features and global contextual information through a spatial-channel attention module, attaining state-of-the-art performance at the time of its release. Zhang et al. [38] improved point cloud segmentation by representing deep features across multilevel structured point clusters, effectively minimizing intraclass distances and maximizing interclass separability. Addressing the lack of robustness in existing methods for complex airborne LiDAR point clouds, Jiang et al. [39] proposed a method that implements local neighborhood feature extraction, global feature representation, and ground-aware attention modules to enhance segmentation outcomes. Zhang et al. [40] introduced the double attention neural network (DA-Net), which incorporates the augmented edge representation (AER) and elevation attentive pooling (EAP) modules, to unveil and efficiently propagate the latent geometric information within point clouds. In advancement to KPConv, Zeng et al. [41] proposed a recurrent residual dual attention network (RRDAN) that achieves high-precision semantic segmentation of airborne LiDAR point clouds by incorporating attention KPConv (AKPConv) and global and local channel attention (GLCA) modules within the encoder and decoder phases, respectively. While these methods demonstrate effective results, they all rely on high-density and superior-quality annotated information.

B. Weakly Supervised Learning in LiDAR Point Cloud

The annotation of large-scale point clouds is time-intensive and costly. Therefore, weakly supervised learning by using sparse annotations becomes a viable approach that achieves desirable results with a minimum of high-precision annotations. For example, using only 0.1% of the labels achieves results that are competitive with those obtained through full supervision learning [10].

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Several studies have investigated weakly supervised learning for airborne LiDAR point clouds. Wang and Yao [42] employed sparse random sampling and pseudo-label generation from predictive results to diminish the reliance on extensive labeling for training. They also investigate the effects of various weak label initialization techniques on weakly supervised learning and introduce a density-biased strategy that outperforms random label initialization methods [43]. Incorporating entropy regularization, consistency constraints, and pseudo-label supervision to furnish multifaceted supervisory signals also improves the robustness of the network [10]. Chen et al. [44] introduced a feature perturbation weakly supervised network (FPWS-Net), incorporating a dual-semantic reasoning structure for enriched point cloud feature learning and amalgamating consistency constraints with a reciprocal pseudo-label loss strategy to bolster the understanding of unlabeled point characteristics. Lin et al. [12] proposed an overlap area loss and an elevation attention module in a classification network to create high-quality pseudo-labels and concurrently implement a supervised contrastive loss in the segmentation network to extract effective features from these pseudo-labels.

Although this way can reduce the annotation workload, for high-precision annotation requirements, it is still necessary to use professional software to label the point cloud or scene one by one, which not only requires higher skills and higher costs but also limits the scale of processing data. Given that the annotations of high-precision point clouds are both time-consuming and labor-intensive, is it really necessary? Conversely, low-precision annotations are cheaper and widely sourced, with methods such as crowdsourcing [19], pseudo labels in weakly supervised learning [12], large-scale annotation generation from existing models [21], and label transfer from 2-D images [23] and so on. It's a promising direction that has not been fully explored, and that's what we'll focus on next.

C. Noisy Learning in Related Fields

Given that noisy labels can have a large impact on the learning of the network and lead to the unusability of large amounts of easily accessible data (e.g., crowdsourced annotations [19]), realizing robust learning with noisy labels becomes an urgent problem to be solved [45]. Currently, most research about learning with noisy labels predominantly occurs within image processing, which learns images with noise labels robustly by choosing the correct label among the noisy labels.

1) Noise Learning in Images: As research on image robust learning with noisy labels commenced early, various methods have been established [45]. Robust learning has been achieved through network structure modifications. Xiao et al. [46] proposed a framework based on probabilistic graphical models for better noisy label correction within an end-to-end framework. Goldberger and Ben-Reuven [47] incorporated a softmax layer to bridge correct and noisy labels, emphasizing label associations. However, this approach bears certain constraints. The incorporation of a robust loss function has also enhanced network learning robustness. Zhang and Sabuncu [48] proposed a noise-robust loss function that can be applied to various frameworks by generalizing the mean absolute error (MAE) and categorical cross entropy (CCE). Wang et al. [49] mitigated network overfitting and underfitting to noisy labels by refining the cross-entropy loss function. In addition, Ma et al. [50] improved the robust learning of the network through the introduction of a robust loss function, yet this method struggles with more complex, noisy labels. One technique for achieving network-robust learning involves reliable label selection. By simultaneously training dual deep neural networks and sifting reliable labels through the process, Han et al. [51] offered a joint pedagogical paradigm. They further improve this method by filtering data with inconsistent predictions while utilizing this data with smaller losses for back propagation [52]. Zheng et al. [53] considered label correction as a dual optimization challenge, utilizing a label correction model to revise incorrect labels and train the primary model, thereby achieving promising results. However, this method risks losing the correct labels and neglecting the potential advantages of noise labels. Renovating labels has also been proven beneficial to network learning robustness. Song et al. [54] advanced a novel noisy label training method called selectively refurbish unclean samples (SELFIE), which limits the impact of noisy labels on the network by prioritizing high-accuracy, corrected noisy labels for training. It effectively utilizes both correct and incorrect labels. However, challenges may arise from incorrect corrections.

In image semantic segmentation with noise labels, considering the spatial relationship of the noisy labels, Yao et al. [55] encoded them by improving the Markov model and proposing a label correction method for images. Liu et al. [21] utilized the phenomenon of noise learning to locate the beginning of network noise memorization and introduce cross-scale consistency operations, successfully suppressing over-fitting of noisy labels. However, the drawback is that it requires high-quality initial labels.

2) Noise Learning in Point Clouds: Point cloud noise learning has not received much attention. Pioneering a robustness study on indoor point clouds with noisy labels, Ye et al. [18] introduced a noise-robust learning method independent of noise rate. They achieved the desired results by obtaining reliable labels by point-confidence selection and correcting the noise for clustering levels. Despite these advances, the study fails to illuminate the specific influence of point cloud noise on network learning and relies too heavily on well-structured data, colored texture information, and high-precision super-voxels to effect satisfactory constraints. Moreover, the differences in scale, complexity, density, and features between indoor and outdoor point cloud scenes make the method unsuitable for learning from airborne LiDAR point clouds with noisy labels.

In this study, we explore for the first time the impact of noise on point cloud segmentation networks. Based on observed phenomena, we introduce a new framework that incorporates modules such as weak category priority, DM, and HC. This framework improves the network's resistance to noise through

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the learning phase, consequently realizing robust learning of airborne LiDAR point clouds with noisy annotation.

III. Method

In this section, we begin by quantitatively analyzing the impact of symmetric noise on network learning in Section III-A. Following that, in Section III-B, we intensively investigate the influence of asymmetric noise on the network, monitoring the learning status of ground truth labels and noisy labels via the metrics $\mathrm{IoU_{gt}}$ and $\mathrm{IoU_{nl}}$. Sections III-C and III-D offer comprehensive details of our proposed modules: the weak class priority (WCP), the DM, and the HC. We then conclude with an overview of our comprehensive framework in Section III-E.

A. Preliminaries

The study of noise learning relies on noise-annotated datasets suitable for quantitative evaluation. There are three types of data currently available for noise learning research: 1) manually annotated datasets with real noise [20], [56]; 2) simulate noise labels based on clean annotations according to specific rules [18], [57]; and 3) simulate noise annotations by pretrained models [21], [58], [59], [60].

Primarily, the annotated dataset containing real noise may consist of certain noise labels. However, the absence of ground truth prohibits quantitative noise analysis. Thus, numerous studies [21], [57] follow the second method for noise simulation. Similarly, for airborne LiDAR point clouds, we can simulate different noise ratios by selecting the same number of points for each category with the furthest point sampling (FPS) algorithm and then flipping the labels of the points by a predefined percentage. Specifically, let the noise rate be $P \in [0, 1]$. We perform symmetric noise operations on the point cloud processed by the FPS algorithm and flip one class to other classes with the same probability $P/(C-1)$ to ensure a uniform distribution of one class into other classes. Fig. 1(a) shows the symmetric noise matrix with a noise rate of 50% on the Large-scale ALS data for Semantic Labeling in Dense Urban Areas (LASDU) dataset [61]. Performances on the LASDU dataset, using noise rates varying from 0% to 50%, are presented in Fig. 1(b). The results demonstrate a negative correlation between noise rate and performance metrics on the LASDU dataset. Specifically, overall accuracy (OA) and mean intersection over union (mIoU) decrease by approximately 12 and 10 percentage points, respectively, at a 50% noise rate. The experimental results show that a certain level of noise labeling can seriously affect the network's learning.

The second simulation method offers the benefit of quantitative noise ratio control. Nonetheless, its assumptions are overly idealized, assuming symmetric noise distribution and uniform sample counts across categories, which diverge notably from actual noise conditions [18], [57]. Therefore, we choose the third method for investigating airborne LiDAR point cloud noise learning. Initially, the model is trained under weak supervision on a clean dataset, followed by evaluation of the dataset to obtain an accuracy assessment dataset reflecting real-world asymmetric noise conditions [21]. Further experiments and analyses focus on the asymmetrically noisy dataset generated using the third method.

B. Learning With Asymmetric Noise

We use the third method to generate an asymmetric noise dataset on the training sets of the LASDU dataset [61] and the ISPRS Vaihingen 3-D Semantic Labeling benchmark (ISPRS) dataset [62] and take the LASDU dataset as an example. The original dataset is referred to as the **accurately labeled dataset**, and the resulting asymmetric dataset is referred to as the **noise dataset**. The labels from the **accurately labeled dataset** are termed "**original clean labels**," and the labels from the **noise dataset** are termed "**original corruption labels**," as illustrated in Fig. 2(a) and (b). To investigate the effect of noise labels on neural network learning, we adopt the quantification approach from Liu et al. [21]. We utilize the **original clean labels** and the **original corruption labels** from the LASDU training set and monitor the network's response to the noise labels using the following two metrics.

1) *Ground Truth Label Learning IoU_{gt}:* We quantify ground truth label learning using the overlap (measured in terms of the intersection over union (IoU) metric) between the outputs and the corresponding ground truth labels, which are shown in Fig. 2(c), denoted by $\mathrm{IoU_{gt}}$.
2) *Noise Label Learning IoU_{nl}:* We quantify noise label learning using the overlap (measured in terms of the IoU metric) between the outputs and the corresponding noise labels, which are shown in Fig. 2(c), denoted by $\mathrm{IoU_{nl}}$.

We train on the processed LASDU noise dataset and show $\mathrm{IoU_{gt}}$, $\mathrm{IoU_{nl}}$, and the IoU of the network's predictions relative to **original corruption labels** in Fig. 3 with respect to the ground, building, tree, and low vegetation categories.

In Fig. 3, the blue line corresponds to $\mathrm{IoU_{gt}}$, the orange-yellow to $\mathrm{IoU_{nl}}$, and the red line denotes the IoU. As can be seen from the curves, there is a downward trend in the $\mathrm{IoU_{gt}}$ across distinct categories, signifying a decrease in the model's memorization of ground truth labels. Simultaneously, $\mathrm{IoU_{nl}}$ shows an upward trend, which indicates increasing memorization of the incorrect labels by the model. Furthermore, it should be noted that the learning efficacy with respect to ground truth labels varies by category, as evidenced by the different peak values of $\mathrm{IoU_{gt}}$ for each category. In certain categories, for instance, low vegetation, the $\mathrm{IoU_{gt}}$ is notably low, suggesting minimal learning of ground truth labels. Conversely, categories exhibiting higher peak values in $\mathrm{IoU_{gt}}$, such as ground, demonstrate superior learning of ground truth labels. We refer to the categories with lower and higher $\mathrm{IoU_{gt}}$ values as "**weak**" and "**strong**" categories, respectively.

We conduct simulations with symmetric noise on both datasets, observing consistent phenomena in each. The experimental phenomena indicate that, throughout the learning process, the network learns ground truth and noise labels almost concurrently. However, proficiency in learning ground truth labels varies among categories. Overall, a consistent pattern emerges across all categories: a decline in the network's ability to learn ground truth labels coupled with an ascendancy in the learning of noise labels.

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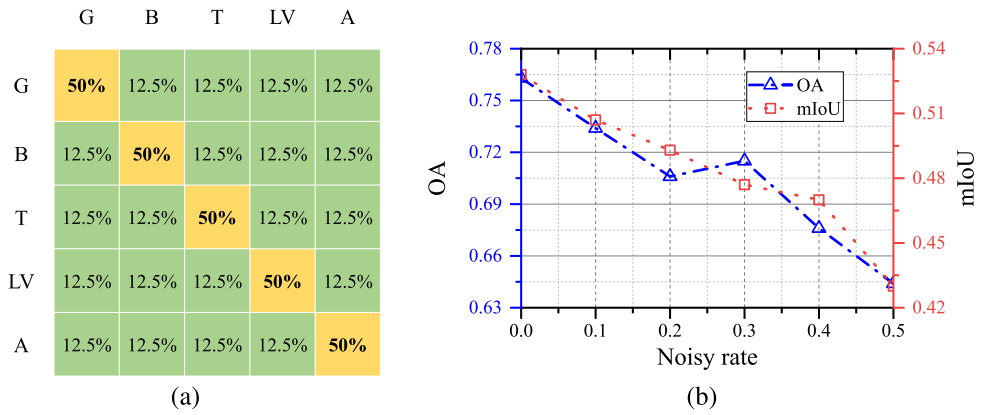

Fig. 1. Noise simulation on the LASDU dataset. (a) Symmetric noise matrix at a 50% noise rate, where G denotes ground, B denotes building, T denotes tree, LV denotes low vegetation, and A denotes artifact. (b) Performance at different noise rates.

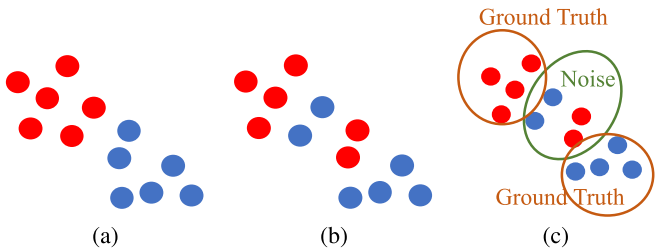

Fig. 2. Brief illustration of an asymmetric noise dataset. (a) Original clean labels. (b) and (c) Original corruption labels.

C. Sequential Adaptive Correction

Based on the experimental phenomena in Section III-B, we consider using the model's predictions to correct noisy labels. This approach has achieved desirable results in the field of image classification [21], [63], [64], [65], [66]. Considering that correcting point cloud noisy labels too early, the network has not yet learned sufficiently on the ground truth labels, and correcting it too late causes the network to learn most of the noisy labels. To determine the best time for label correction, the following factors need to be considered.

1) The learning efficacy of distinct categories on the ground truth labels varies, as indicated by the different peak values in $\mathrm{IoU_{gt}}$.
2) When training on real datasets, there is no prior knowledge such as the noise rate to observe how the network learns the ground truth labels and the noisy labels. Therefore, the only thing we can utilize is the correlation between the network's predictions and the **original corruption labels**, such as the IoU.
3) The network learns faster for strong categories than for weak categories. As a result, the weak category takes longer to reach a higher IoU. If the labels of strong categories are corrected earlier than those of weak categories, it inevitably leads to much better results for the strong categories than the weak ones.

To handle these challenges, we introduce a WCP module and a DM module. For each category, the WCP module initially distinguishes between strong and weak categories by analyzing the IoU between the network's prediction and the **original corruption labels** [as shown in Fig. 2(b)]. For weak categories, we use the DM module to detect the IoU between the network prediction and the **original corruption labels** to locate the appropriate time to correct these labels. Subsequently, we prioritize the updating of weak categories. When prioritizing the update of strong categories, the segmentation network learns more features about these categories and therefore recognizes them better and better. However, weak categories require longer learning for the segmentation network to recognize them. Prioritizing the updating of the strong categories leads the network to focus more on these categories, which inevitably makes the network fail in building a well-structured feature space for weak categories. The WCP module ensures that weak categories learn more ground truth labels at an earlier stage, thereby minimizing the potential impact of noisy label corrections in strong categories. For instance, compare a house composed of millions of points with a utility pole composed of 10 000 points. The network learns the former's features more effectively, reaching the appropriate label correction point earlier. If we apply label correction to the house first, the network will be more efficient in learning this category. However, the network will struggle to learn the features of the 10 000-level utility pole. Ultimately, subsequent to the correction of labels within weak categories, corrections are concurrently implemented across both strong and weak category labels.

1) Weak Category Priority Module: In the experiment, we find that the IoU curve between the network's prediction and the **original corruption labels** [as shown in Fig. 2(b)] has a particular association with the peak value of $\mathrm{IoU_{gt}}$. When the peak value of $\mathrm{IoU_{gt}}$ is low, the IoU between the network prediction and the **original corruption labels** often learns slower and with a lower peak value. Motivated by these observations, we introduce a threshold $P(0 < P < 1)$. At the e_ith epoch, the IoU of the network's prediction and the **original corruption labels** are assessed to differentiate between strong categories (characterized by a higher peak value) and weak categories (identified by a lower peak value).

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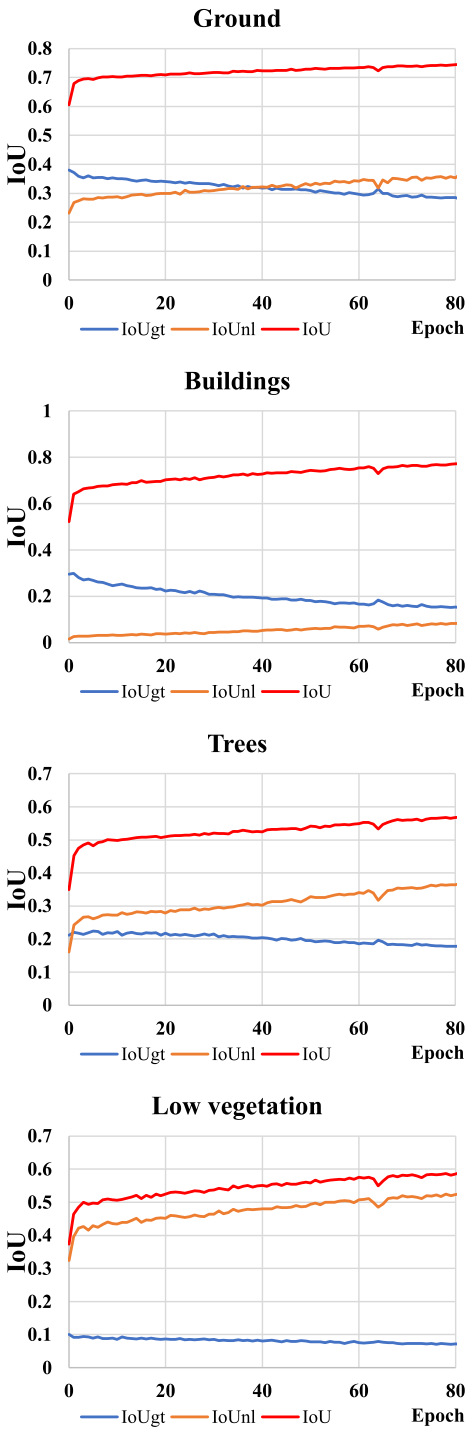

Fig. 3. We visualize the changes of the IoU (red), IoU_{gt} (blue), and IoU_{nl} (orange) curves on the LASDU training set.

The formula (1) is as follows:

$$\text{Rate}_c = \begin{cases} \text{Strong}, & \text{IoU}_c >= P \\ \text{Weak}, & \text{IoU}_c < P \end{cases} \tag{1}$$

where c represents the category, IoU_c represents the IoU of the network prediction and the **original corruption labels** for this category, and Rate_c represents the strength attribute of this category. We set P as 0.5 and e_i as 2.

2) DM Module: In order to detect the changes in the IoU curve for each category, we use the following ordinary least square (OLS) model to fit the IoU curve of the training set by the IoU values of adjacent multiple epochs. The formula (2) is as follows:

$$Q = \min \sum_{i}^{n} (\hat{y}_i - y_i)^2 \tag{2}$$

where Q represents the error, $\hat{y}_i$ represents the estimated result, and y_i represents the ground true value. In the context of a fit binary linear model, label correction commences at a specific epoch when the slope K is ascertained to be below the threshold $K_{\max}$, and simultaneously, the IoU exceeds $\text{IoU}_{\min}$.

D. Select Labels in History

The HC module is designed to correct noisy labels. Initially, the correction time within weak categories is determined by applying the WCP module and the DM module. Subsequently, the HC module selects reliable labels derived from the network's prediction to correct the noisy labels.

We represent point clouds as $\mathbf{X} \in \Gamma^{(N \times F)}$, where N denotes the number of points, F denotes the features of points, such as the number of echoes, Y represents the semantic label, and C represents the class number. The predicted label for the nth point at the ith epoch within the model is expressed as $h_{in}, n \in \{1, \ldots, N\}$. The model stores historical predicted labels as $H_l = \{h_{1N}, \ldots, h_{iN}\}$, where l denotes the history length (e.g., usually set to 4). For a point $P_n, n \in \{1, \ldots, N\}$ possessing l historical labels, its frequency belonging to class c,$c \in \{1, \ldots, C\}$ is calculated and denoted as $P_n(c)$

$$P_n(c) = \sum_{h_{in} \in H_l} [h_{in} = c] \tag{3}$$

where the square brackets [.] represent the Iverson brackets. Subsequently, the label $c_n^{\max}$, which is the most recurrent in the historical predicted labels H_l for the point P_n, is computed

$$c_n^{\max} = \arg\max_{x_i} P_n(x_i), \quad x_i \in \{1, \ldots, C\}. \tag{4}$$

For each point P_n, compute the probability of occurrence for the label value $c_n^{\max}$. This probability is symbolized as $P_n^{\max}$

$$P_n^{\max} = \frac{\sum_{h_{in} \in H_l} [h_{in} = c_n^{\max}]}{l}. \tag{5}$$

The confidence level Y_n for each point P_n is computed through the application of entropy

$$Y_n = \frac{1}{-\log_{\frac{1}{C}}} \left(-P_n^{\max} \log P_n^{\max}\right), \quad n \in (1, \ldots, N). \tag{6}$$

The confidence level Y_n for each point P_n is compared with a predefined threshold θ (e.g., usually set to 0.001). Labels that manifest a confidence level below θ are deemed reliable, denoted as Yrel_n, and are thus retained for subsequent network learning. Conversely, labels surpassing the confidence

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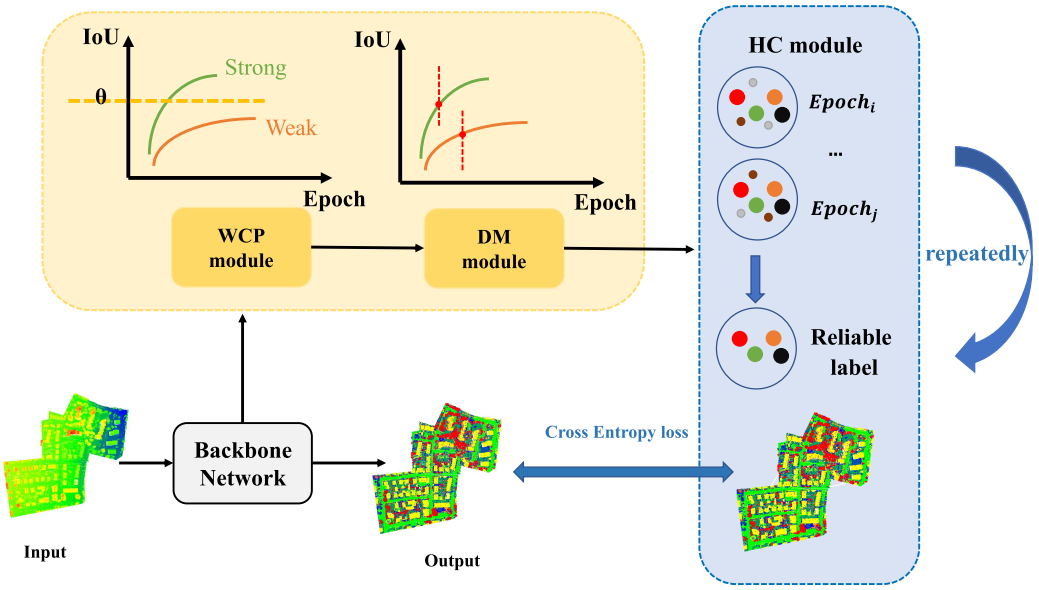

Fig. 4. Overall framework. First, the WCP module classifies strong and weak classes based on the network's learning of ground truth labels. Then, the DM module is utilized to identify the moments when the weak class has sufficiently learned the ground truth labels while not overfitting the wrong labels. At this point, the HC module is applied to correct these noisy labels. This module is used repeatedly throughout the training process to completely replace noisy labels.

threshold θ are classified as unreliable and consequently ignored (e.g., set to -1)

$$\text{Yrel}_n = \begin{cases} c_n^{\max}, & \text{if } Y_n <= \theta, \quad n \in (1, \ldots, N) \\ -1, & \text{others.} \end{cases} \tag{7}$$

Ultimately, we replace the **original corruption labels** with the reliable label set $X_{\text{reliable}} = \{\text{Yrel}_n, n \in (1, \ldots, N)\}$, thus enabling the network to learn robustly against noise labels.

Noisy labels are replaced at regular intervals (epochs). When noisy labels are replaced for the first time, training the network on a cleaner dataset enhances its recognition ability, leading to more accurate predictions. Through repeated replacement, the dataset will become cleaner, and the network's recognition ability will be improved. Ultimately, a stable state is reached between the two processes, effectively maximizing the role of the HC module.

In quantifying the deviation between the network predictions and the ground truth labels, we use the cross-entropy loss function to calculate the loss

$$\text{Loss}_{\text{seg}} = -\frac{1}{P_w} \sum_{i}^{N} \sum_{j}^{C} y_{ij} \log p_{ij} \tag{8}$$

where P_w represents the number of samples within a mini-batch, y_{ij} and p_{ij} denote the labels and predicted results of the point cloud, respectively.

E. Overall Framework

The overall pipeline, primarily composed of the WCP module, the DM module (highlighted in Section III-C), and the HC module (discussed in Section III-D), is shown in Fig. 4. We adopt KPConv [27] as the main framework. In the early stages of training, we classify categories into strong and weak classes based on the relationship between the network's prediction and the **original corruption labels**. This approach guarantees that weak categories are prioritized to learn more ground true labels, avoiding interference from the correction of noisy labels for strong categories in weak categories. Subsequently, the DM module is utilized to determine the optimal timing for correcting labels in weak categories. At this moment, the model has sufficiently learned ground true labels without overfitting noisy labels. Ultimately, we apply the HC module to correct noisy labels by selecting reliable labels from a multitude of network predictions. The core idea behind our approach is to leverage the network's predictive outcomes at appropriate timing to discern reliable labels, subsequently replacing the noisy labels for robust learning.

IV. Experiment

A. Datasets

To validate the efficacy of our method, we perform experiments with two distinct airborne LiDAR point cloud datasets: the ISPRS [62] and the LASDU [61].

1) ISPRS Vaihingen Dataset: The dataset is airborne LiDAR point clouds acquired by Leica ALS50 system scanning in Vaihingen, Germany. It exhibits a point density of approximately 6.7 points/m^2. Each point within the dataset contains information on XYZ-coordinates, intensity, number of echoes, and total number of echoes. The dataset is partitioned into two distinct segments: a training set, containing 753 859 points, and a test set, containing 411 721 points. It encompasses a total of nine categories, namely, powerline, low vegetation, impervious surfaces, car, fence/hedge, roof, facade, shrub, and tree. The visualization of the point cloud along the Z-axis is shown in Fig. 5. We set

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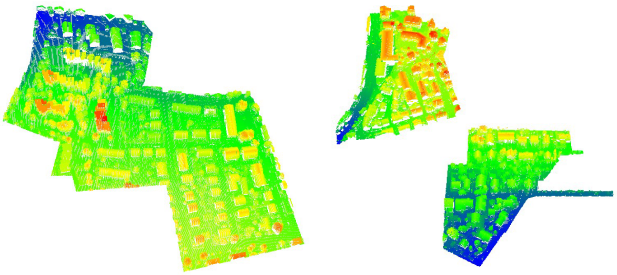
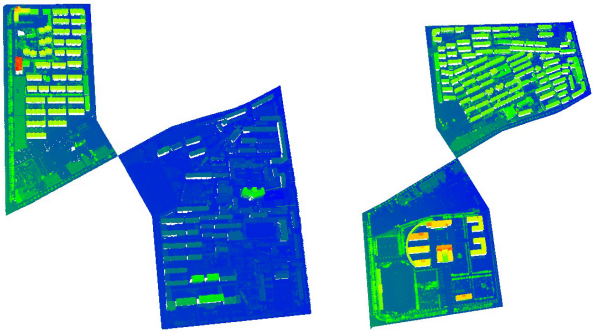


(a) (b)

Fig. 5. ISPRS dataset. (a) Training set. (b) Test set.



(a) (b)

Fig. 6. LASDU dataset. (a) Training set. (b) Test set.

the downsampling grid size to 0.4 m and utilize all feature information for network training.

2) LASDU Dataset: The airborne LiDAR point cloud dataset is acquired by the Leica ALS70 system over the Heihe River Basin, China. Comprising approximately 3.12 million points, the dataset exhibits an average point density of roughly 4 points/m^2. It is partitioned into four segments, wherein the training set includes approximately 590 000 points across two segments and the test set includes approximately 1.13 million points across two segments. There are five different categories included in this dataset: ground, artifacts, low vegetation, trees, and buildings. The rendering effect of the point cloud along the Z-axis is illustrated in Fig. 6. We set the down-sampling grid size to 0.6 m and utilize the XYZ-coordinates and intensity information for network training.

We follow [21], [58], [59], and [60] to generate asymmetric noise labels for the ISPRS and LASDU datasets. The corresponding confusion matrices are shown in Fig. 7(a) and (b), respectively.

B. Implementation Details

In our experiments, all models are implemented in the framework of PyTorch 1.8.2 and trained on an NVIDIA GeForce RTX 3090 24-GB GPU. We adopt the KPConv network as the backbone for our models. For the ISPRS dataset, we configure the radius to 20 m, while for the LASDU dataset, it is set to 40 m. During each epoch, we set the batch size to 8 and the number of steps to 120 for the ISPRS dataset. Conversely, for the LASDU dataset, the batch size is adjusted to 4, with 200 steps per epoch. We impose a maximum of 400 epochs for training on both datasets. Throughout the training process, we employ the stochastic gradient descent (SGD) optimizer to refine the network parameters. The initial learning rate is established at $1e^{-2}$, with a momentum value of 0.98. All other parameters are kept consistent with the default settings of the KPConv segmentation network. We set the threshold P for delineating strong and weak classes to 0.5 for both datasets. In addition, we dynamically identify the IoU curve using a linear equation.

C. Evaluation Metrics

We evaluate the performance of our method using OA and mIoU. OA represents the proportion of data correctly classified on the test dataset. mIoU represents the average IoU for each class on the test dataset. The expression is as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{9}$$

$$\mathrm{mIoU} = \frac{1}{C} \sum_{i=1}^{C} \frac{\mathrm{TP}}{\mathrm{FN} + \mathrm{FP} + \mathrm{TP}} \tag{10}$$

where TP, TN, FP, and FN, respectively, represent true positives, true negatives, false positives, and false negatives in a confusion matrix.

D. Experiments and Analyses

To evaluate the effectiveness of our method, we perform experimental evaluations on the ISPRS 3-D labeling Vaihingen and LASDU datasets, comparing our results against various established methods. Given the scarcity of research on robust outdoor point cloud noise learning, we compare our method with point noise-adaptive learning (PNAL) [18], which studies noise learning in point clouds and has demonstrated favorable outcomes in robustness against noise labels in indoor scenes. The method utilizes per-point confidence selection to identify historically reliable labels for each point and leverages clustering information to perform label correction within the most similar local regions. It applies the density-based spatial clustering of applications with noise (DBSCAN) algorithm [67], which uses XYZ-coordinates and RGB color information, to cluster indoor point clouds with some success. When clustering airborne LiDAR point clouds, the absence of color information and uneven density distributions in certain regions might limit the effectiveness of DBSCAN. In the processing of DBSCAN, the algorithm is more prone to overlooking small-volume objects like "cars" in low-density areas compared to large, densely distributed categories such as "ground." To mitigate this issue, we employ the super voxel method to construct clustering information, which can identify compact and uniform local regions with similar features. Meanwhile, we compare the image classification noise label robustness learning method to combine the generalized cross entropy loss (GCE) [48] into the KPConv framework. Finally, we compare it with the adaptive early-learning correction (ADELE) method [21], which represents the state-of-the-art approach in the field of image semantic segmentation. During training, it detects the onset stage of noise memory for each class and adaptively corrects the noisy labels using enforced cross-scale consistency. Considering its primary focus on image processing, applying the same up-sampling

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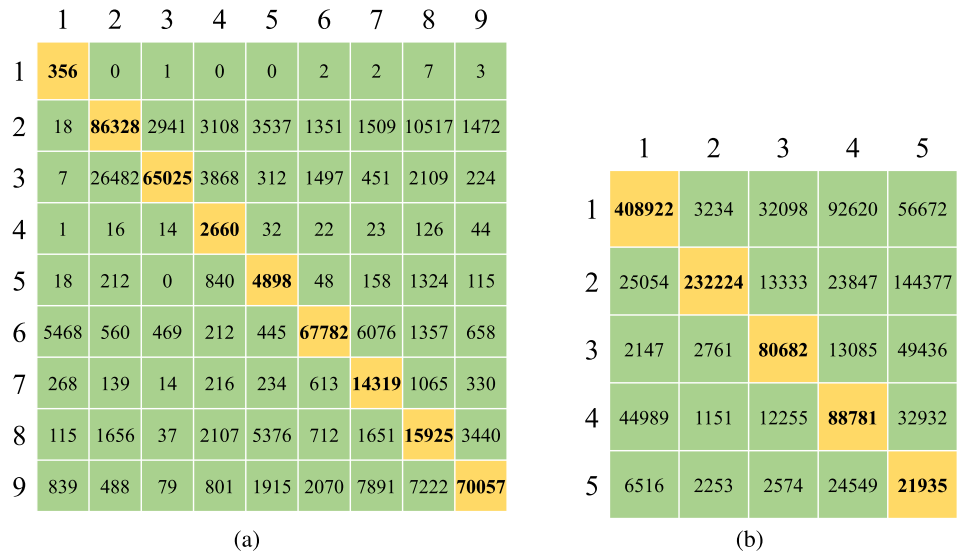

Fig. 7. Confusion matrix for noise-label and ground truth label. (a) ISPRS dataset. (b) LASDU dataset.

TABLE I

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE ISPRS DATASET, WHERE FULL SUP. INDICATES DIRECT TRAINING WITH CLEAN LABELS AND DTNL INDICATES DIRECT TRAINING WITH NOISY LABELS

Method	IoU									OA	mIoU
	1-Power	2-Low veg	3-Imp.Surf	4-Car	5-Fence	6-Roof	7-Facade	8-Shrub	9-Tree		
Full Sup.	0.606	0.670	0.820	0.627	0.175	0.882	0.486	0.261	0.690	0.821	0.580
DTNL	0.158	0.536	0.673	0.202	0.146	0.773	0.376	0.228	0.577	0.709	0.408
GCE [48]	0.000	0.550	0.710	0.369	0.145	0.808	0.380	0.224	0.567	0.730	0.417
ADELE [21]	0.245	0.550	0.762	0.186	0.148	0.823	0.352	0.215	0.587	0.734	0.430
PNAL [18]	0.000	0.551	0.747	0.226	0.101	0.766	0.167	0.151	0.563	0.729	0.364
Ours	**0.547**	**0.633**	**0.774**	**0.527**	**0.182**	**0.831**	**0.417**	**0.242**	**0.625**	**0.782**	**0.531**

and down-sampling techniques directly to the point cloud may degrade the quality of the data and prevent the network from learning the various classes correctly. Therefore, we choose to replace the cross-scale consistency method in the study with the HC module.

In our experiment, only noisy labels from the training set are utilized, excluding the use of ground truth labels. Tests are conducted on test sets with ground true labels that have not undergone any noise processing. The performance results for the two datasets are presented below.

1) Results on ISPRS Dataset: Table I presents the comparison between fully supervised learning, direct training with noisy labels (DTNL), GCE, ADELE, PNAL, and our method on the ISPRS dataset. GCE shows no improvement over DTNL and cannot recognize the power category. This demonstrates that only modifying the loss function is not an effective way in LiDAR point cloud noise learning. The ADELE approach produces a slight improvement of about two points in the results, which is due to its focus on early learning stages without considering the mutual influence between categories. From the results, it can be seen that PNAL leads to a deterioration in performance. Specifically, the powerline category fails to be learned, and there is a notable decrease in the IoU for both the facade and shrub categories. This is because the greater complexity of airborne

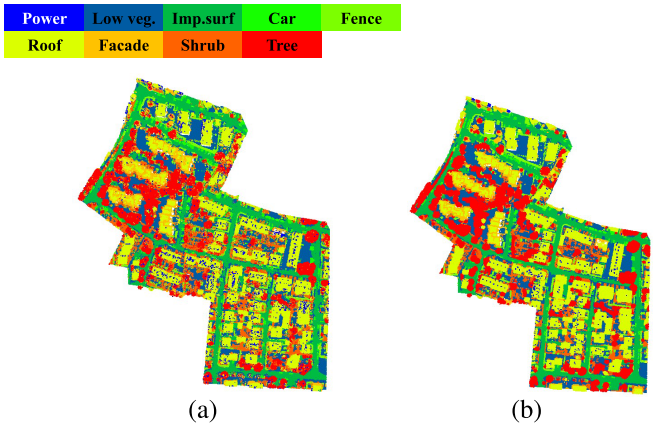

Fig. 8. Overall results on the ISPRS dataset. (a) Direct training with noise labels. (b) Our method.

LiDAR point clouds with categories like vegetation and trees hinders the acquisition of precise cluster-level information in contrast to indoor point clouds. As a result, applying the cluster-level correction from the PNAL approach to airborne LiDAR point clouds is ineffective. Our approach yields a 12-point improvement in IoU compared to the baseline direct training with the noisy labels method. All categories have experienced enhancements, notably the power category, which

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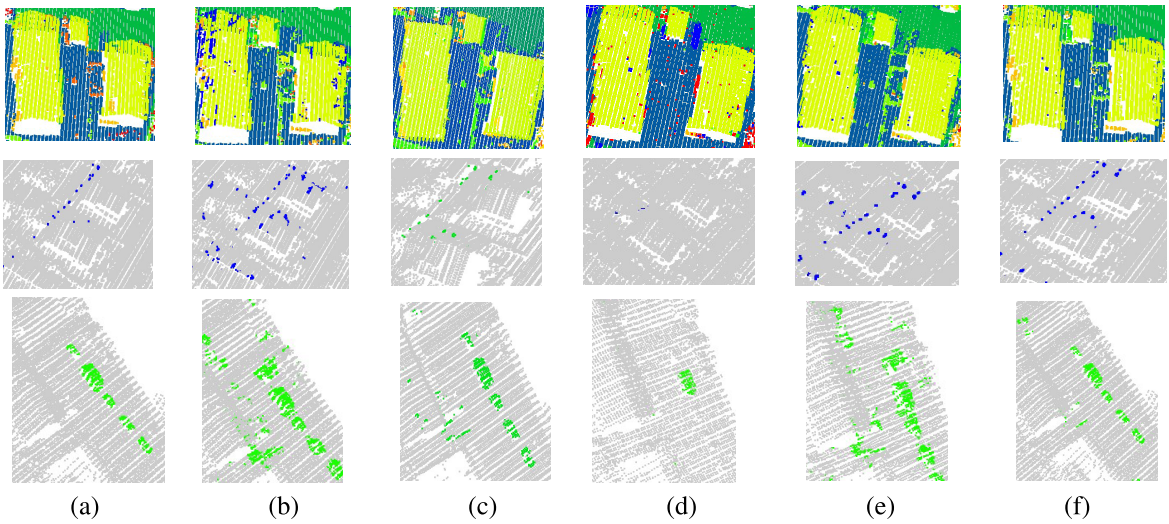

Fig. 9. Detailed results on the ISPRS dataset. (a) Ground truth. (b) Direct training with noise labels. (c) GCE [48]. (d) PNAL [18]. (e) ADELE [21]. (f) Our method.

has witnessed a substantial IoU increase of approximately 38 points. This improvement is attributed to the considerable influence of our WCP module. By diminishing the impact of easy-to-learn categories such as low vegetation, impervious surfaces, and powerlines, we facilitate the effective learning of this category from the ground truth labels. The improvement observed in the fence and shrub categories is inconsequential. This is due to the limitations of fully supervised learning, where the IoUs of the two categories are only 17.5% and 26.1%, respectively. This shows that the network struggles to recognize them. The network's ability to recognize these categories will become more difficult when noise is present in these categories, which will also lead to a decrease in prediction accuracy. Using these predictions to correct the labels does little to improve the network's recognition of these categories.

The overall segmentation results are also presented in Fig. 8. The results show that the noise labels for categories such as buildings and facades have been significantly reduced by our method compared to the baseline. For a comprehensive demonstration of our method's effectiveness, we visualize the segmentation results of some regions in Fig. 9. The top three rows correspond sequentially to the categories of building, powerline, and car. In the case of the powerline category, GCE [48] and PNAL [18] are unable to accurately identify this category. For the GCE, it cannot eliminate the effect of noise on the powerline category. For the PNAL, this is because the powerline's cluster information contains other categories, causing the PNAL's cluster correction to be inefficient.

While the ADELE method [21] learns the powerline category, it still captures a considerable amount of noise. This is because ADELE does not adequately perform label substitution, resulting in a lot of noise remaining in the dataset. In contrast to their approach, ours considers the effects between different categories and chooses to update the noise labels multiple times using reliable labels. As a result, this strategy enhances performance across many different categories (as shown in the last column).

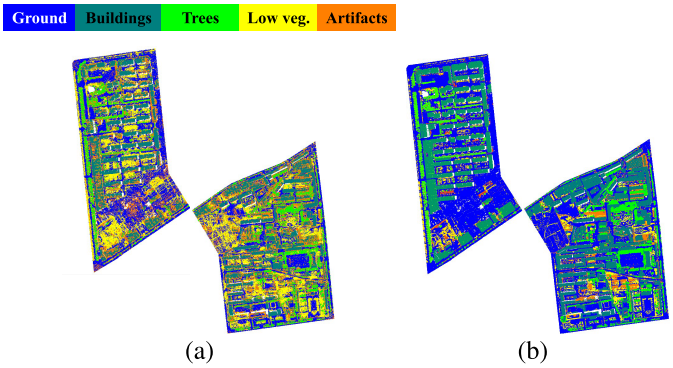

Fig. 10. Overall results on the LASDU dataset. (a) Direct training with noise labels. (b) Our method.

2) Results on LASDU Dataset: Table II presents the comparison among fully supervised learning, direct training with noisy labels, GCE, ADELE, PNAL, and our method on the LASDU dataset. Compared with DTNL, the performance of GCE in some categories such as ground, tree, and artifact is degraded, which indicates that the effect of noisy labels on the network is not well suppressed by only modifying the loss function. The results indicate that the ADELE method does not show a large improvement overall, only in the building and low vegetation categories, while the other categories show a decrease. The shortcoming of the ADELE is that it does not consider the interrelationships between the categories, resulting in different rates of progress in label correction, with categories such as buildings progressing at a faster rate than categories such as artifacts. As learning progresses, the network improves for the building category but deteriorates for the artifact category. PNAL does not yield favorable results due to the complexity of airborne LiDAR point clouds. This complexity hinders the formation of pure clusters for each class, leading to the presence of mixed-class clusters. Therefore, PNAL's cluster-level correction of airborne LiDAR point clouds is unsuccessful. Compared to direct training with noisy labels, our approach enhances the IoU by approximately

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TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE LASDU DATASET, WHERE FULL SUP. INDICATES DIRECT TRAINING WITH CLEAN LABELS AND DTNL INDICATES DIRECT TRAINING WITH NOISY LABELS

Method	IoU					OA	mIoU
	1-Ground	2-Buildings	3-Trees	4-Low Veg	5-Artifacts		
Full Sup.	0.815	0.919	0.763	0.384	0.296	0.858	0.635
DTNL	0.745	0.739	0.541	0.200	0.084	0.724	0.462
GCE [48]	0.730	0.741	0.486	**0.240**	0.078	0.722	0.455
ADELE [21]	0.728	0.796	0.533	0.217	0.058	0.749	0.466
PNAL [18]	0.728	0.766	0.541	0.215	0.045	0.753	0.459
Ours	**0.762**	**0.887**	**0.734**	0.202	**0.092**	**0.805**	**0.535**

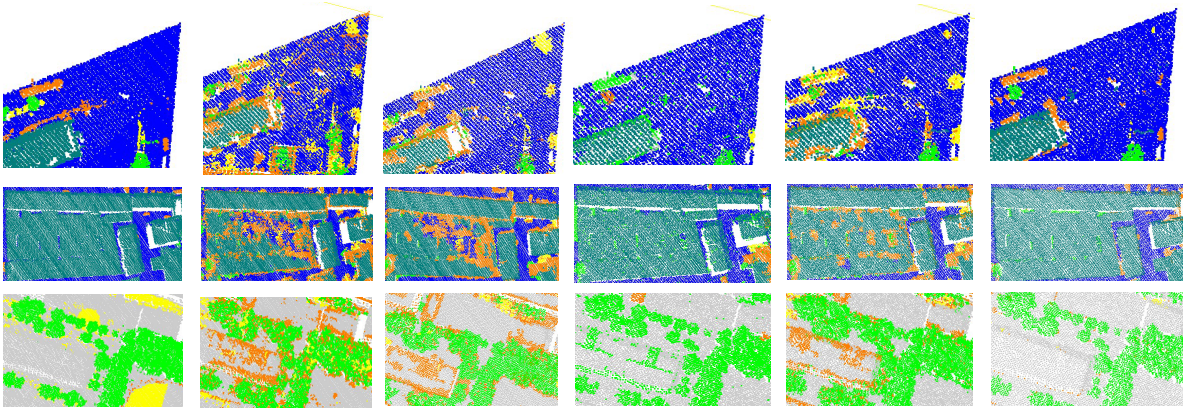

Fig. 11. Detailed results on the LASDU dataset. (a) Ground truth. (b) Direct training with noise labels. (c) GCE [48]. (d) PNAL [18]. (e) ADELE [21]. (f) Our method.

7 percentage points overall and noticeably increases the building category by approximately 15 percentage points.

We present the overall segmentation results on the LASDU dataset in Fig. 10. It is evident that for categories including ground, buildings, and trees, we have successfully mitigated the impact of noise and accomplished considerably desirable outcomes. We visualize the segmentation results in more detail, as shown in Fig. 11. The top three rows, in descending order, correspond to the categories of ground, building, and tree, respectively. For the ground and building categories, our method (last column) effectively reduces noise interference, enabling the network to learn their features more effectively. However, GCE [48], PNAL [18], and ADELE [21] methods still contain a significant amount of noise. In the third row, GCE [48] does not remove the effects of noise, resulting in that trees still containing a lot of noise. Simply modifying the loss function is not an efficient way to do this. Same in the third row, it is apparent that PNAL [18] erroneously misclassifies buildings as the tree category. This is because PNAL performs cluster-level correction on outdoor airborne LiDAR point clouds, and the similarities in height and other properties between the building, tree, and artifact categories inevitably result in the wrong categories being clustered together. Therefore, correcting clusters with different categories to the same category leads to lower results. The ADELE [21] incorrectly identifies the building category as the artifacts category in the third row because it fails to consider the interference of the artifacts category caused by the building category. Conversely, our method considers intercategory interference and accurately learns tree category features, thus clearly distinguishing the tree category from the building category. Our approach requires that the **original corruption labels** of a category contain a certain percentage of correct labels so that the network can learn features from the correct labels and correct the noisy labels of that category. For the ISPRS dataset, an appropriate proportion of correct labels exists for all categories, so that the network learns the features of all categories, and the noisy labels are replaced for all of these categories, resulting in a large increase in mIoU. For the LASDU dataset, a low percentage of correct labels exist for some categories, resulting in the network not being able to learn the features of these categories and therefore not being able to modify the noisy labels of these categories. So the mIoU enhancement is less compared to the ISPRS dataset.

E. Ablation Study

In this section, we conduct ablation experiments on each module. The results are shown in Table III. This part of the experiment is also performed on the ISPRS and LASDU datasets processed by the third noise simulation method. The data in Table III indicate that, for the ISPRS dataset, the implementation of the DM and HC modules results in an increase of approximately 6% in OA and an enhancement of roughly 9% in mIoU compared to the baseline model. Nevertheless, when compared to the outcomes achieved by

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TABLE III
EFFECTIVENESS OF WCP, DM, HC ON DIFFERENT DATASETS

Module			ISPRS		LASDU	
WCP	DM	HC	OA	mIoU	OA	mIoU
			0.709	0.408	0.724	0.462
	✓	✓	0.773	0.498	0.765	0.515
✓		✓	0.772	0.511	0.767	0.501
✓	✓		0.671	0.411	0.712	0.445
✓	✓	✓	**0.782**	**0.531**	**0.805**	**0.535**

employing the full complement of modules, the mIoU is observed to be 3% lower. Regarding the LASDU dataset, the application of the DM and HC modules leads to an approximate enhancement of 4% in OA and a 5% rise in mIoU relative to the baseline model. However, when compared with the results derived from the full module application, the mIoU remains 2% inferior. The analysis of both datasets reveals that the DM and HC modules substantially enhance the network's robustness against noise and proficiently amend erroneous labels. Similarly, the WCP and HC modules show similarly positive results. The WCP and DM modules do not yield any improvements in either dataset. This is because, with the WCP and DM modules, the network simply determines the starting point for noise memory, but no noise label correction is performed, resulting in no improvement.

V. CONCLUSION

In this article, we explore the noise learning of airborne LiDAR point clouds. We present an asymmetric noise simulation approach for airborne LiDAR point clouds and conduct an extensive analysis of model performance against noisy labels. Then, we introduce a novel framework adaptive dynamic noise label correction (ADNLC), which is specifically engineered to facilitate robust learning in the presence of noise labels. In ADNLC, the WCP module is used to assist the network in perceiving strong and weak classes, and the DM module is used to find moments when the network has fully learned the ground truth labels without overfitting the wrong labels. The HC module is developed to replace wrong labels by selecting reliable labels from historical predictions. We conduct comprehensive experiments on two benchmark ALS datasets. Compared to other methods, our framework shows significant improvements in terms of OA and mIoU. Ultimately, we establish a new baseline for noise learning in airborne LiDAR point clouds.

Although our approach achieves good results, there are some limitations. We only consider the historical consistency of the point labels, not the structural information or topological relationships of the objects, which are to be investigated later. In addition, we will further explore how to leverage noisy labels from remote sensing images or crowdsourced OSM [20]. As large-scale models and multimodal learning flourish, cross-modal noise labels will become increasingly accessible. This trend brings unprecedented opportunities and challenges for point cloud noise learning. We believe that in the near future, point cloud noise learning will usher in greater development potential by effectively utilizing these rich data sources.

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